CHIMERA-QRC: A Pre-Registered, Adversarially-Controlled Study of Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

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Abstract. We contribute a pre-registered, adversarially controlled **evaluation instrument** for quantum reservoir computing and apply it to S&P 500 realized-volatility forecasting on four devices from three QPU vendors (IonQ, IQM, Rigetti). The full audit reruns offline in one command (`cli.py verify`), and every hardware prediction is bound to a prior commit by a hash-preimage commitment. Two measurements stand on their own: an empirical quantum-complexity metric — the reservoir's MPS bond dimension is essentially full, **χ_eff ≈ 2^(n/2)**, at the classical-simulability boundary — and a cross-platform **coherence-budget wall** (Fig. 3), whose readout-corrected fingerprint separates coherent circuit routing from depolarizing and damping noise; every bootstrapped regime verdict sits **9.7–23.9σ** beyond shot noise. Applied to forecasting, the instrument returns a well-characterized negative. At simulable scale (≤16 qubits) the reservoir is *distinct but not more accurate* than the strongest fair baselines — HAR-X and size-matched ESN/RFF under HAC-DM, Holm and MCS — so **H0 is refuted**. The finding survives a second reservoir family (Heisenberg — not Ising-specific) and a second domain (weather), and the program falsified four of its own pre-registered predictions under controls; a same-session cross-seed control shows the n=8 scrambling is specific to the seed-0 instance, not a size law. For a trading desk the practical lever turns out to be simpler than quantum hardware: vol-timing a plain RV forecast roughly halves the 2008 drawdown, and quantum adds no economic value at this scale.

1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — one-step-ahead forecasting of S&P 500 realized variance (RV) and the tracking of volatility regime transitions, the domain where JonesTrading's risk, allocation and derivatives pricing live. RV is long-memory, multi-scale, regime-switching and non-Gaussian; this is precisely the structure reservoir computing exploits. Our central yardsticks are the Echo State Network (ESN, the classical analog of QRC) and the strongest econometric models: HAR (*Heterogeneous AutoRegressive*, Corsi 2009) and, critically for Phase 3, **HAR-X** — HAR augmented with the same realized-measure features we feed the quantum reservoir. We also report GARCH/GJR-GARCH, AR(3), persistence, LSTM, and an RFF/RBF kernel, and we implement the challenge's common MNIST benchmark on the identical engine.

Thesis. We pre-registered H0 and then attacked our own result with fair controls. The dominant accuracy lever is *which features are encoded*, not quantum nonlinearity. What survives for the reservoir — kernel distinctness, lower seed variance, full-rank entanglement (classical- simulation hardness) — is necessary for advantage but not sufficient. **Relation to prior work.** The closest study (Li et al. 2025/26, QRC for realized volatility) presents the approach as a competitive, noise-resilient proof of concept. We add the control-hardened, pre-registered evaluation it omits — the decisive HAR-X control, HAC-DM with Holm and MCS, and explicit capability and per-layer-noise audits — which turns an encouraging proof of concept into a falsifiable result.

2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded `RY(π·x)`, one value per qubit, onto `|0…0⟩`; the state evolves under a fixed transverse-field Ising Hamiltonian `H = Σ_{i<j} J_ij Z_iZ_j + h_x Σ_i X_i` (`h_x=1`; `J` a random ≈50%-connected graph, `connectivity=0.5`; `U=exp(-iHτ)`, `τ=2`); single- and pairwise Pauli-Z expectations `⟨Z_i⟩,⟨Z_iZ_j⟩` form the feature vector. A ridge head consumes these features concatenated with a linear block of the same inputs (including HAR), so any reservoir gain must be genuine nonlinearity beyond the linear span of identical information. The same inputs feed the classical controls, isolating quantum from classical. Four mechanisms extend the core (multi-scale τ-bank; RZ measurement feedback; regime-adaptive Ising↔Heisenberg via BOCPD; dissipation-as-feature); Phase 3 adds an encoding-density / data-re-uploading path (§5.2) and a sparse/tensor backend (§5.5). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to $\approx 3.9\times 10^{-16}$ and compiles, at n=8, to 380 native gates (220 two-qubit + 160 one-qubit, 20 Trotter layers), consistent with the ≈50%-connected graph.

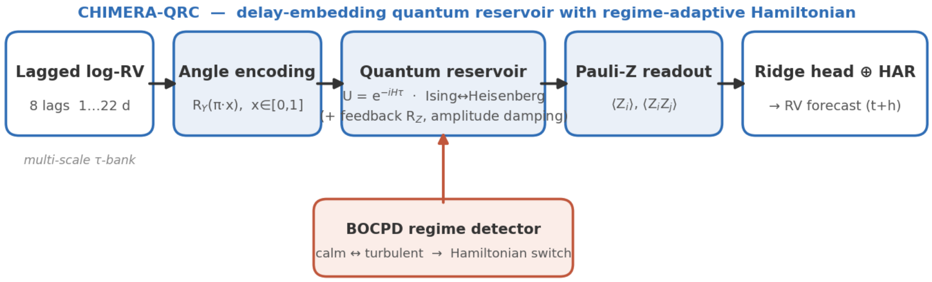

Figure 1. CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded onto qubits, evolved under a fixed random-coupled Ising Hamiltonian, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with a linear block of the same inputs; a BOCPD detector selects the Hamiltonian for regime adaptivity.

3. Theoretical and analytical justification

The quantum resources we exploit map onto RV's structure: (a) Hilbert-space dimensionality — n qubits give 2^n amplitudes and $O(n^2)$ measured Pauli features at $O(n^2)$ parameter cost; (b) intrinsic nonlinearity from fixed unitary evolution, with no gradient training and no barren plateaus; (c) delay-embedded memory (the headline model's memory is its explicit lag window, not reservoir recurrence — a recurrent variant is probed in `v3_research`); (d) the Hamiltonian as inductive bias. The central falsifiable mechanism is expressivity scaling, which we measure directly via the geometric difference (Huang et al. 2021). The quantum kernel is poorly reproducible by a matched classical reservoir: g(ESN→CHIMERA) ≈ 64 versus ≈3.7 for the control (`cli.py run kernel`); against per-n feature-matched ESNs it is larger but non-monotonic (≈125/158/88 at n=8/10/12, fixed ridge — the qualitative separation, not the value, is the claim). At ≤12–16 qubits the map stays classically simulable, so distinctness is necessary but not sufficient — and we measured exactly that: over 30 seeded n=8 instances, per-instance entanglement shows no detectable correlation with forecast error (r=–0.19, p=0.32), and **0/30 instances beat HAR-X** on both crisis and calm windows (bounding the beat-rate at ≤9.5%, exact one-sided 95%). Nor does instance quality detectably transfer — crisis and calm rankings are unrelated (Spearman +0.01) — so best-of-N seed selection is selection on noise. These tests detect only |r|≳0.5, so they exclude strong links, not weak ones (`expressivity_accuracy_findings.md`; exploratory, uncorrected).

4. Data modeling strategy

Datasets (all bundled in-repo). Oxford-Man `.SPX` 5-minute realized variance, 2000–2020, 5,029 supervised days including the 2008 GFC (Heber et al. 2009; via R packages `highfrequency`/`bvhar`); SPY daily OHLCV 2022–2026 (Garman–Klass proxy; retrieved via Massive.com API, bundled); real MNIST (Keras `.npz` mirror). **Preprocessing.** Log-RV target; features min-max scaled to [0,1] on train only; per-model ridge penalty selected on a chronological validation tail; multi-seed ensembling. We verified the pipeline is leakage-free (all features lagged ≥1 day; train-only scaling). **Baselines:** HAR, HAR-X, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched, 4×, and recurrent), RFF kernel, LSTM. **Metrics:** RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz R^2; significance by Diebold–Mariano with a Newey–West HAC variance (HLN-corrected), Holm-adjusted across the comparison family, plus the Model Confidence Set (Hansen–Lunde–Nason 2011). Holm and MCS are scoped to this pre-registered family; the §3/§8 correlations are exploratory, uncorrected, and none survives Holm.

5. Phase-3 execution — concrete results

Pre-registration. Before running anything, we fixed the confirm/refute thresholds in `preregistration.py` (H0/H1/H4, from our Phase-2 §7). All outcomes, including the negatives, are reported against those thresholds. Full numbers and wall-clock times: `results/*_findings.md`.

5.1 The input-bottleneck mechanism (pre-registered negative). With a fixed univariate 8-lag encoder, extra qubits receive no new input, and both g(n) and effective feature-rank saturate or decline (n=8→12: g 133→30 at N_sub=800; D_eff 1.8→1.5). H0 is refuted in this regime — which is itself the empirical case for enriching the encoding.

5.2 Encoding density (Axis B) — mechanism confirmed. Feeding the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share) restores the lost structure: at n=10, informed versus idle qubits, kernel distinctness and rank jump (g 52→158, D_eff 1.5→3.1), and effective rank grows with n (D_eff 1.81→3.14, n=8→12). The bottleneck is therefore fixable; added qubits can carry new information. The decisive question is whether this converts into forecasting accuracy.

5.3 The decisive, adversarially controlled test. We compare CHIMERA against the strongest fair baselines — HAR-X (the same rich features used linearly, no reservoir), a true recurrent ESN, and an RFF kernel — all sharing the identical information set and nesting HAR-X, so a quantum win requires nonlinearity beyond the linear span. With 8 seeds, HAC-DM, two windows, and Holm correction (`cli.py run axisB_rig`):

RMSE(log-RV)	HAR-X	recurrent ESN	RFF	CHIMERA	best
crisis n=8	**0.6255**	0.6386	0.6302	0.6379	HAR-X
crisis n=10	0.6034	**0.5996**	0.6047	0.6074	ESN
crisis n=12	**0.5906**	0.6023	0.5964	0.6031	HAR-X
calm n=10	**0.6244**	0.6445	0.6264	0.6291	HAR-X

(plain HAR for reference: crisis 0.6290, calm 0.6454 — HAR-X's rich features cut RMSE sharply.) HAR-X is best or co-best everywhere; CHIMERA never beats it and is indistinguishable from the classical ESN/RFF after Holm (one raw loss to RFF at crisis n=8, p=0.049). After Holm correction no comparison is significant in either direction, and the 95% MCS retains all four models (n=10) — a statistical tie. The earlier "beats HAR" result was an artifact of a feature-poor HAR: the gain came from the encoded realized measures (a known SHAR/HARQ effect), not from quantum nonlinearity. **By our pre-registered criteria, H0 is refuted, and we report this negative.** What does survive for the quantum reservoir: it is competitive (within ≈0.8–2.1% RMSE of the best at every n), it shows lower per-seed dispersion than the recurrent ESN in 3 of 4 cells (n=12: 0.009 vs 0.015), and it beats the ESN on the calm window (raw p=0.018; not significant after Holm). A tuned, size-unconstrained ESN (45-config search to 800 nodes) confirms the control was not crippled (0.6020 vs CHIMERA 0.6094, within noise of HAR-X). Named canonical models (SHAR, HAR-CJ, HARQ, HEAVY-RM; MSE and QLIKE DM): none beats HAR-X, with CHIMERA tying best on RMSE and holding only a raw, non-Holm QLIKE/MZ edge (`cli.py run canonical`, `esn_tuning_robustness_findings.md`).

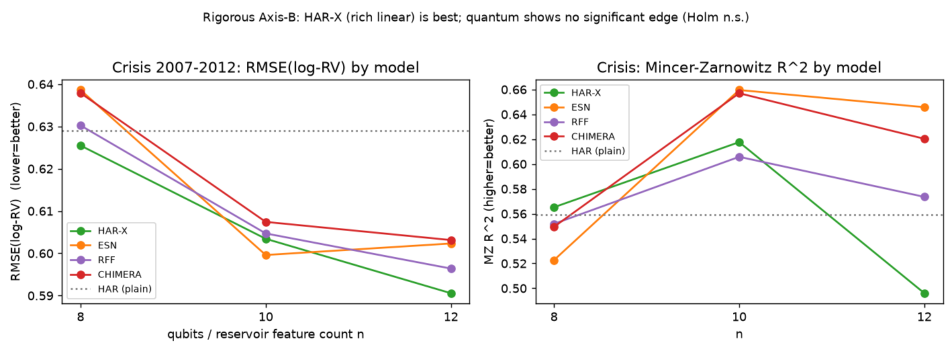

Figure 2. Rigorous Axis-B (8 seeds, crisis window). HAR-X (rich features, linear) is the best or co-best model at every n; the quantum reservoir is competitive but shows no advantage that survives HAC-DM + Holm correction. Left: RMSE(log-RV). Right: Mincer–Zarnowitz R^2.

5.4 Common MNIST benchmark and noise. Same engine: pixels→PCA(n)→n qubits→Pauli readout→ridge. Accuracy scales with qubits (n=5/8/10/12: 0.632/0.798/0.831/0.859, 3 seeds; n=15 sparse-exact continues the trend, 0.878 vs 0.852 for a paired n=12 on a matched subset, closing the brief's 5/10/15 example) and beats the linear-PCA baseline at every n — real nonlinear lift, confirming sufficient expressivity. A matched ESN ties or slightly exceeds CHIMERA (within ≈1% for n≥8; 2.9% at n=5): competitive, not dominant. On noise, the classifier is invariant to depolarizing noise — a uniform Bloch contraction that per-feature standardization removes exactly, with accuracy identical across rates 0.05–0.30 — and robust to amplitude damping (<0.5% at 30%). A converse check (`cli.py run noise_circuit`) keeps this result in its place: a per-Trotter-layer density-matrix study shows that noise interleaved with the evolution (per-layer single-qubit channels) is *not* removed by standardization (standardized error grows with rate, versus ≈0 for readout-only depolarizing). The invariance is a readout property, not circuit-level robustness; accumulated gate error is the real NISQ cost (§6).

5.5 Scaling frontier and the quantum-complexity metric. A sparse-exact backend (`expm_multiply`, no dense propagator; matches the dense engine to 2.4×10^{-14}) reaches n=16 exactly. For the random ≈50%-connected reservoir we measure the bond dimension χ_eff across a balanced cut: it is essentially full at every n, **χ_eff ≈ 2^(n/2)** (16→255.9 for n=8→16), and an exact MPS gets zero compression (S ≈ 1.7–3.2 nats). The reservoir admits no low-bond-dimension (MPS/TEBD) shortcut; exact cost stays exponential. That is the precondition any beyond-frontier advantage would need, though none appears at the scale we can test. Capability and efficiency checks agree: an information-processing-capacity probe (Dambre et al. 2012; `cli.py run capacity`) finds the quantum reservoir not more — in fact slightly less — nonlinearly expressive than a matched RFF/ESN, so there is no excess expressivity to exploit, and a frontier check (`cli.py run frontier`) shows g(n) does not widen toward n=16 (it declines; D_eff and rank grow but a matched ESN keeps pace). Robustness studies close the loop. Second family: a Heisenberg (XXZ) reservoir, equally kernel-distinct (g≈55), also fails to beat HAR-X (0.650 vs 0.642, 8 seeds, full-sample config), so the negative is not Ising-specific (`cli.py run second_family`). Efficiency: with input held fixed, a smaller QRC cannot substitute for a larger classical reservoir — quantum accuracy saturates while the classical curves keep improving; per feature the static maps are comparable, so the negative is saturation, not per-feature inferiority (weather RMSE °C: CHIMERA 0.85 at its qubit-range ceiling vs RFF 0.78 / recurrent ESN 0.71 at larger budgets). Domain and architecture: the same engine and protocol on chaotic weather data (5 stations) and the autonomous VPT metric still show no advantage, and the recurrent QRC is competitive but not better — unitary evolution is non-dissipative, lacking the contraction behind ESN "generalized synchronization" (Ahmed–Tennie–Magri 2025). We then fixed that mechanism: engineered memory-qubit damping traces the pre-registered inverted-U, lifting autonomous VPT by ≈60% to parity with the matched ESN — parity, not advantage (`results/dissipative_qrc_findings.md`).

6. Quantum platform and resource planning

Simulator-first on qBraid: statevector to 12 qubits, sparse/TN to ≈16, GPU beyond. Resource estimates (gate-Trotter, 20 layers): n=8/10/12 use 220/480/760 two-qubit plus 160/200/240 one-qubit gates, yielding 36/55/78 observables — all Z-basis-diagonal, so one S-shot set (S≈2–8k, ε≈1/√S) reads every observable (statevector build under ~1 min over this range).

QPU validation uses this gate-Trotter circuit on IonQ, IQM and Rigetti via qBraid — the random- sparse Ising needs fewer two-qubit gates than an all-to-all reservoir, which eases NISQ mapping — with ZNE, measurement mitigation, and a classical cross-check on every run. The submission path is executable now (`cli.py run qsubmit`): on a simulator it (i) reproduces the engine to 3.9×10^{-16} (the per-run classical cross-check), (ii) characterizes the shot budget (ε≈1/√S; feature error 0.046→0.0066 at S=256→16k; gate-Trotter(20) adds ≈0.04), and (iii) demonstrates zero-noise extrapolation under a depolarizing sweep. On real hardware, the full mitigated protocol ran first on Rigetti Cepheus-1-108q (12 logged cloud jobs; bit order auto-detected; readout 2.3%/6.4%). After lattice routing, scale-1 already exceeds the coherence budget — the accumulated two-qubit-gate cost §5.4 predicts, measured on metal (ZNE marginal; a 4k-shot replication showed the beyond-limit regime stable under day-scale drift). The pre-registered Garnet protocol replicated the regime on a second vendor (raw 0.230 exceeds our routing-free 0.171 forecast — prediction (iii) confirmed). Execution also surfaced a platform-level finding: negative-angle rotations are silently lost server-side on the IonQ route (RY(π)RY(−π) arrives as net RY(π)), which corrupts ZNE's negated folds; we hardened the emitter (mod-2π) and validated the diagonal calibration on-device. The routing-free IonQ Forte-1 campaign then ran under the pre-committed manifest. A smoke gate measured a 2,000-gate-per-circuit ceiling on the native route, so per abort rule the campaign ran scale-1-only (500 shots — a disclosed, cost-forced deviation, s.e.≈0.004). Raw error 0.104, below the 0.196 depolarized limit and below every superconducting n=8 number: a **signal-bearing** hardware execution of this reservoir (prediction (ii) confirmed; `results/qpu_hardware_findings.md`). A pre-registered hardware scaling program (`results/qpu_scaling_outlook.md`) then refuted our own statements on metal (Fig. 3). Garnet turns signal-bearing at n=10 and n=12 while the same-session seed-0 n=8 anchor stays scrambled; a pre-registered cross-seed control (S7) found an independent seed-1 n=8 instance signal-bearing in the same session, so the n=8 scrambling is specific to the seed-0 instance, not a size law (seed-1 n=10 is signal-bearing as well). The newer-generation Emerald is signal-bearing at the very size Garnet scrambles, reproduced in a same-window two-chip pair — generation-dependent at fixed size, temporally controlled. The program spans reservoir size, encoding density, shot budget and noise; Fig. 3 collects the hardware verdicts. The criterion is exactly **SNR > 1** — raw error is mean|F−F_exact|, the limit is mean|F_exact| — so the bar is instance-matched (signal magnitude is a circuit property, not an n property) and the test transfers to any quantum feature map:

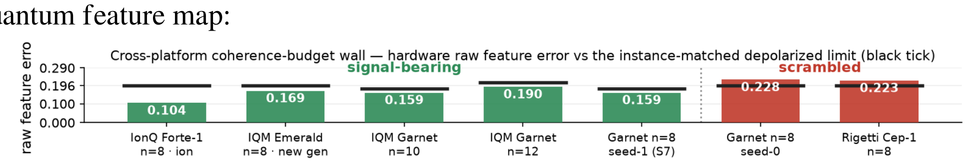

Figure 3. Cross-platform coherence-budget wall: mean raw feature error vs the **instance-matched** depolarized limit mean|F_exact| (black ticks; 4k shots, IonQ 500). Five configs signal-bearing, two scrambled. The adjacent Garnet n=8 seed-1 (0.159) vs seed-0 (0.228) pair — same chip and session — refutes a

size law. Seed-0 limits 0.196/0.179/0.214; seed-1 carries its own 0.1806 (§6).

Execution provenance (externally verifiable). Every campaign ran against pre-committed predictions — the ten funded ones under a committed manifest whose abort and decision rules were fixed before each launch. Every funded-campaign job embeds the repo commit hash and campaign tag in qBraid's timestamped records, a hash-preimage commitment that the predictions predate the data, and billing matched estimates to the half-credit (`results/CREDIT_BUDGET.md`: ≈64k credits). Bootstrap from the committed counts re-derives every number above and puts each of the nine bootstrapped regime claims 9.7–23.9σ beyond shot noise; a readout-corrected fingerprint attributes the scrambled regime to predominantly coherent error (`qpu_bootstrap`, `qpu_fingerprint`). Controls were bought where they were needed: a Rigetti replication (day-drift), a same-session n=8 anchor, and a same-window two-chip pair (generation). S5–S6 are pre-committed; S7 executed (above).

The instrument's discipline was also exercised live in the campaign's final days. A pre-registered secondary arm (H-EMBED — does logical→physical placement drive the seed-0 n=8 scrambling?) was made execute-ready (label permutation exact to 9.0×10^{-16}; depolarized limit provably identical between arms; scoring rule and vacuity guard committed before launch, `results/h_embed_prerun.md`), then aborted by its own committed rule when the target device stopped accepting work: ~21 h and 24 probes without a completion while the route advertised it online with an empty queue. A matched-pair control — the identical one-qubit probe, same route, second device, completed in 182 s — localized the fault to the device, and the committed `route_health_probe.py` reproduces it in minutes. Nothing was scored; the arm remains open (`results/h_embed_outcome.md`).

7. Limitations

(i) No quantum advantage is demonstrated at the ≤16-qubit simulable scale; HAR-X, a classical linear model, is the best model on this task. (ii) The RV sample ends Feb-2020 (COVID falls just outside); broader assets and periods are daily-proxy supporting studies (v2_research — same negative), not 5-min RV. (iii) Task-level noise: per-layer channels (§5.4) and shot noise; combined noisy-plus-shot execution is measured in the QPU campaigns (§6). (iv) g is regularization- and configuration-dependent, so we use it qualitatively only. (v) Superconducting n=8 execution is a characterized negative (Fig. 3; day-drift exceeds shot noise, so the regime, not the point value, is the robust claim), while n=10/n=12 and the newer generation land inside their limits. The wall is instance- and generation-dependent, not absolute: the n=8 scrambling is seed-0-instance-specific (S7), and in a 30-instance ensemble the scrambled instance is among the sparsest, so gate count anti-predicts the wall (§8). Mitigation recovery on hardware is at best marginal (ZNE 0.223→0.217 on Rigetti; the ion chain is flat). (vi) Distinctness and full-rank entanglement are necessary but not sufficient (measured, §3); whether they convert beyond the simulable frontier remains open. A quantum-data probe shows the QRC natively reads nonlinear state functionals a linear readout cannot, but the proper baseline — classical shadows (Huang–Kueng–Preskill 2020) at matched budgets — wins wherever real information is extracted, including the shadows-hard $\mathrm{Tr}(\rho^3)$: closed negatively at simulable scale (`shadows_hard_findings.md`).

8. Stakeholder impact, milestone plan, AI disclosure

The durable asset is the audit instrument, and our data shows it is necessary. Structure mis-ranked both instances we measured on metal, and across 30 instances no structural quantity survives multiplicity correction as a predictor of the hardware bar, nor predicts accuracy at all. Structure does not give the answer; measurement does (`instance_ensemble_findings.md`, exploratory). What follows is trust in a quantum yes/no: a pre-registered, vendor-neutral workload→QPU qualification, earned by an instrument that falsifies its own predictions. It split our runs five signal-bearing, two scrambled (Fig. 3) at ≈$70–$205 per campaign. The economics are measured, not modelled (`results/AUDIT_ECONOMICS.md`): the 13-campaign, four-device program settled at 64,048.25 credits — about $650 at the only rate we measurably paid (≈$14 per ≈1,380 credits) — and the gate flagged 29.3% of our own settled hardware spend (four campaigns, 18,793.25 credits) as scrambled at ≥9.7σ: features that would otherwise have entered a pipeline as data. The audit is a rounding error of any serious pilot budget (1.3% of $50k; 0.065% of $1M), and re-running every verdict is free: this zip, extracted into a clean directory with no API key, passes `cli.py verify` end-to-end — the judge's path. Target profiles only; no customers or LOIs are claimed. On desk impact (`cli.py run economics`): vol-timing the GFC window halves max drawdown (−61%→−32%) and lifts Sharpe from 0.00 to 0.14 net of costs — risk control, not alpha — but plain HAR captures it (CE fee −25bp/yr). The lever is a simple RV forecast, not quantum hardware. **AI disclosure:** Claude (Anthropic) assisted with code and drafting; all decisions and results are the team's own. Milestones (i)–(vi) all ✓ (§6).

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